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ASHRAE Research: Improving the Quality of Life

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ASHRAE Research: Improving the Quality of Life

ASHRAE is the world's foremost technical society in the fields of heating, ventilation, air conditioning, and refrigeration. Its members worldwide are individuals who share ideas, identify needs, support research, and write the industry's standards for testing and practice. The result is that engineers are better able to keep indoor environments safe and productive while protecting and preserving the outdoors for generations to come.

One of the ways that ASHRAE supports its members' and industry's need for information is through ASHRAE Research. Thousands of individuals and companies support ASHRAE Research annually, enabling ASHRAE to report new data about material

properties and building physics and to promote the application of innovative technologies.

Chapters in the ASHRAE Handbook are updated through the experience of members of ASHRAE Technical Committees and through results of ASHRAE Research reported at ASHRAE conferences and published in ASHRAE special publications, *ASHRAE Transactions*, and ASHRAE's journal of archival research, *Science and Technology for the Built Environment*.

For information about ASHRAE Research or to become a member, contact ASHRAE, 1791 Tullie Circle N.E., Atlanta, GA 30329; telephone: 404-636-8400; www.ashrae.org.

Preface

The 2020 ASHRAE Handbook—HVAC Systems and Equipment discusses various systems and the equipment (components or assemblies) they comprise, and describes features and differences. This information helps system designers and operators in selecting and using equipment. ASHRAE Technical Committees in each subject area have reviewed all chapters and revised them as needed for current technology and practice.

Some of the volume's revisions and additions are as follows:

- **Chapter 7**, Combined Heat and Power Systems, has a new section on economic evaluation and includes an update on EU Directive 2004/8/EC.
- **Chapter 9**, Applied Heat Pump and Heat Recovery Systems, has new content on waste heat recovery, district applications, and industrial process heat pumps.
- **Chapter 12**, District Heating and Cooling, has new content from ASHRAE research project RP-1267 (the new *District Heating Guide* and *District Cooling Guide*).
- **Chapter 18**, Variable Refrigerant Flow, has new sections on modeling and system commissioning, and an updated system design example.
- **Chapter 19**, Duct Construction, has extensive revisions on system leakage and air dispersion systems.
- **Chapter 20**, Room Air Distribution Equipment, has updates for current technology, with new information on specialized components and air curtains.
- **Chapter 21**, Fans, has new sections on series fan operation and field performance testing plus added content on fan and motor efficiency grades and parallel multiple-fan operation.
- **Chapter 24**, Desiccant Dehumidification and Pressure-Drying Equipment, has expanded content on applications, air filters, and liquid strainers, plus recommendations from ASHRAE research project RP-1339 on rating equipment at altitude.
- **Chapter 25**, Mechanical Dehumidifiers and Related Components, has new content on psychrometrics, outdoor air, controls, and industrial dehumidifiers.
- **Chapter 26**, Air-to-Air Energy Recovery Equipment, has new information on heat pipes and desiccant and heat wheel systems.
- **Chapter 28**, Unit Ventilators, Unit Heaters, and Makeup Air Units, has revisions on standards, controls, and fan selection for makeup air units.
- **Chapter 29**, Air Cleaners for Particulate Contaminants, has updates on standards and performance testing.
- **Chapter 31**, Automatic Fuel-Burning Systems, has added content on pneumatically and electronically linked gas/air ratio burner systems.

- **Chapter 33**, Furnaces, has updates for current technology and efficiency requirements.
- **Chapter 37**, Solar Energy Equipment, has new data on worldwide solar technology use, plus an expanded section on photovoltaic equipment.
- **Chapter 38**, Compressors, has revisions on general theory; screw and scroll compressors; and bearings, including oil-free technologies.
- **Chapter 44**, Centrifugal Pumps, has new content on vertical, inline, split-coupled pumps; hydronic system pump selection; and differential pressure control.
- **Chapter 45**, Motors, Motor Controls, and Variable-Frequency Drives, has new content on standards, bearing currents, and permanent-magnet motors.
- **Chapter 47**, Valves, has new content on control valve sizing; electronic actuators; and ball, butterfly, flow-limiting, and pressure-independent control valves.
- **Chapter 49**, Unitary Air Conditioners and Heat Pumps, has a new map of U.S. regional appliance efficiency standards.
- **Chapter 50**, Room Air Conditioners and Packaged Terminal Air Conditioners, has updates for efficiency standards.
- **Chapter 52**, Dedicated Outdoor Air Systems, is a completely new chapter focusing on DOAS equipment.

This volume is published, as a bound print volume and in electronic format as a PDF download and online, in two editions: one using inch-pound (I-P) units of measurement, the other using the International System of Units (SI).

Corrections to the 2017, 2018, and 2019 Handbook volumes can be found on the ASHRAE website at www.ashrae.org and in the Additions and Corrections section of this volume. Corrections for this volume will be listed in subsequent volumes and on the ASHRAE website.

Reader comments are enthusiastically invited. To suggest improvements for a chapter, **please comment using the form on the ASHRAE website** or, using the cutout page(s) at the end of this volume's index, write to Handbook Editor, ASHRAE, 1791 Tullie Circle N.E., Atlanta, GA 30329, or fax 678-539-2187, or e-mail hkennedy@ashrae.org.

Heather E. Kennedy
Editor

- Seasonal start-up date
- Occupant move-in date
- Operator training
- Supporting a process, such as operation of computer equipment
- Promoting a germ-free environment
- Increasing marketability of rental spaces
- Increasing net rental income
- Increasing property saleability
- Public image of the property
- Certification of energy use (LEED®, ENERGYSTAR®, etc.)

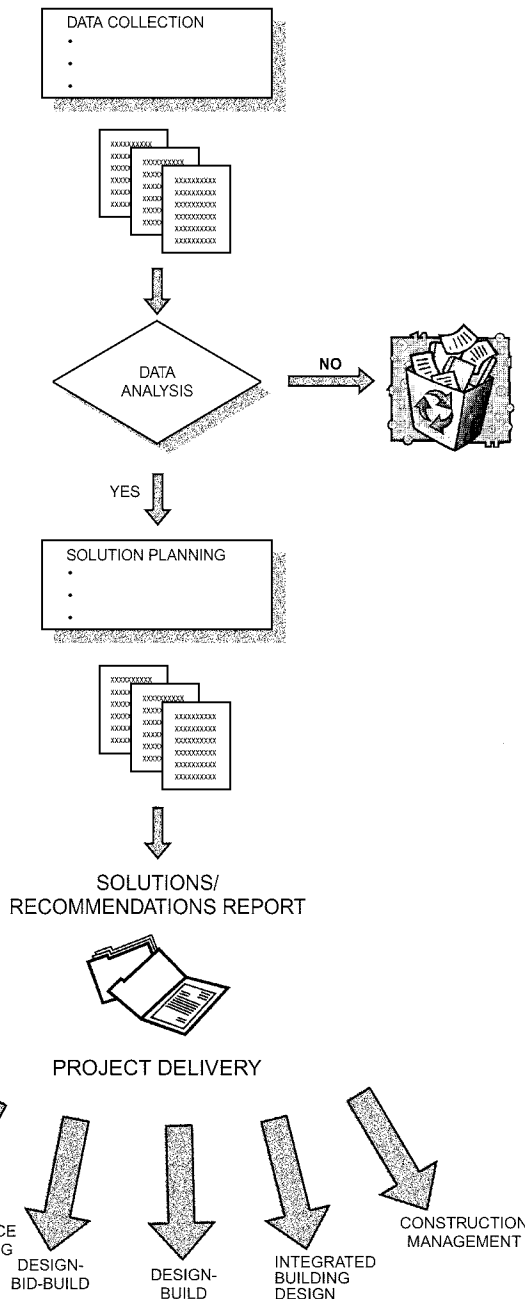


Fig. 1 Process Flow Diagram
(Courtesy RDK Engineers)

The owner can only make appropriate value judgments if the design engineer provides complete information on the advantages and disadvantages of each system option. Just as the owner does not usually know the relative advantages and disadvantages of different HVAC systems, the design engineer rarely knows all the owner's financial and functional goals. Hence, the owner must be proactive and involved in system selection in the conceptual phase of the job. The same can be said for operator participation so that the final design intent is sustainable and can continue to fulfill design intent.

All owners should request and/or require the design team to provide building and HVAC security (as defined in Chapter 61 of the 2019 *ASHRAE Handbook—HVAC Applications*). This security, in addition to environmental health and safety criteria, should address outside-the-building influences (e.g., smoke, toxic fumes, flood, etc.) as well as influences inside the building. System selection pertaining to HVAC security may require another design team member, a security consultant, and/or the owner's own security group.

Equipment and System Constraints

Once the goal criteria and additional goal options are verified, equipment and system constraints must be assessed and documented as part of the system analysis and selection recommendation report. These constraints may include the following:

- Performance limitations (e.g., temperature, humidity, space pressure, etc.)
- Code updates and/or new codes
- Available capacity including equipment size, as well as ductwork and/or pipe size
- Available space and access in and/or out with new or old equipment
- Available utility source
- Energy budget (e.g., conceptual Btu/h-ft² per year), code-driven or targeted
- Equipment efficiency
- Operator knowledge and capabilities
- Existing building occupants
- Building architecture

Few projects allow detailed quantitative evaluation of all alternatives. Common sense, historical data, and subjective experience can be used to narrow choices to one or two potential systems. Heating and air-conditioning loads often contribute to constraints, narrowing the choice to systems that fit in available space and are compatible with building architecture. Chapters 17 and 18 of the 2017 *ASHRAE Handbook—Fundamentals* describe methods to determine the size and characteristics of heating and air-conditioning loads. By establishing the capacity requirement, equipment size can be determined, and the choice may be narrowed to those systems that work well on projects within the required size range.

Loads vary over time based on occupied and unoccupied periods, changes in weather, type of occupancy, activities, internal loads, and solar exposure. Each space with a different use and/or exposure may require its own control zone to maintain space comfort. Some areas with special requirements (e.g., dehumidification requirements) may need individual systems. The extent of zoning, degree of control required in each zone, and space required for individual zones also narrow system choices.

No matter how efficiently a particular system operates or how economical it is to install, it can only be considered if it (1) maintains the desired building space environment within an acceptable tolerance under expected conditions and occupant activities and (2) physically fits into, on, or adjacent to the building without causing objectionable occupancy conditions.

Cooling, heating, and humidity control are often the basis of sizing HVAC components and subsystems, but ventilation requirements can also significantly impact system sizing. For example, if

large quantities of outdoor air are required for ventilation or to replace air exhausted from the building, the design engineer may only need to consider systems that transport and effectively condition those large outdoor air volumes, thus eliminating many of the other HVAC systems (e.g., unit ventilators, through-the-wall window units) from system consideration. The cooling, heating, humidification, and dehumidification delivery performance compromises may be minor for one application in a given climate, but may be unacceptable in another that has more stringent requirements.

HVAC systems and associated distribution systems often occupy a significant amount of space. Major components may also require special support from the structure. The size and appearance of terminal devices (e.g., grilles, registers, diffusers, fan-coil units, radiant panels, chilled beams) affect architectural design because they are visible in the occupied space, which constrains the design.

Sustainable energy consumption can be compromised and long-term project success lost if building operators are not trained to efficiently and effectively operate and maintain the building systems. For projects in which the design engineer used some form of energy software simulation, the resultant data should be passed on to the building owner so that goals and expectations can be measured and benchmarked against actual system performance. Even though the HVAC designer's work may be complete after system commissioning and turnover to the owner, continuous acceptable performance is expected and will only occur if the owner understands the background and systems selected and their anticipated performance. See ASHRAE *Guideline 0* and to ASHRAE's Building Energy Quotient (bEQ) program (www.buildingeq.com).

System operability should be considered in system selection. Constructing a highly sophisticated, complex HVAC system in a building where maintenance personnel lack the required skills for maintenance can be a recipe for disaster at worst, and at best may require the use of costly outside maintenance and service contractors to achieve successful system operation.

The design engineer should closely coordinate the system constraints with the rest of the design team, as well as the owner, to overcome design obstacles associated with the HVAC systems under consideration for the project.

Constructability Constraints

The design engineer must take into account HVAC system constructability issues before the project reaches the construction document phase. Some of these constraints may significantly affect the success of the design and cannot be overlooked in the conceptual and design development phases. Some issues and concerns associated with constructability are

- Existing conditions (e.g., floor load, access into and through a building)
- Rigging equipment into and out of a building to a designated area
- Demolition and impact on adjacent space and existing systems in operation
- Maintaining existing building occupancy, use of the building, and system operation
- Ability to phase HVAC system installation
- Temporary HVAC
- Equipment availability (e.g., delivery lead time)
- Construction schedule
- Construction budget

It can be very advantageous to involve a builder into the early stages of an HVAC design to bring their prospective and experience to the project system analysis and selection process. Whether a construction manager, design-builder, or integrated project delivery leader, each brings years of experience to the topic of constructability that can influence the final selection of HVAC system(s).

Construction budget constraints can also influence the choice of HVAC systems. Based on historical data, some systems may not be economically viable within the budget limitations of an owner's building program. In addition, annual maintenance and operating budget (utilities, labor, and materials) can be limiting although more often than not, annual operating budgets do not get included in the total construction "soft costs" (e.g., furnishing, general conditions). This can be particularly important for building owners who will retain the building for a substantial number of years. The building's estimator can assist in the overall project cost by introducing value-engineered solutions that may involve (1) cost-driven performance, which may provide a better solution for lower first cost; (2) a more sustainable solution over the life of the equipment; or (3) best value based on a reasonable return on investment.

Narrowing the Choices

The following chapters in this volume present information to help the design engineer narrow the choices of HVAC systems:

- **Chapter 2** focuses on a distributed approach to HVAC.
- **Chapter 3** provides guidance for large equipment centrally located in or adjacent to a building.
- **Chapters 4 to 18** address the numerous types of air-conditioning and heating systems.

Each chapter summarizes positive and negative features of various systems. Comparing the criteria, other factors and constraints, and their relative importance usually identifies one or two systems that best satisfy project goals. In making choices, notes should be kept on all systems considered and the reasons for eliminating those that are unacceptable.

Each selection may require combining a primary system with a secondary (or distribution) system. The primary system converts energy derived from fuel or electricity to produce a heating and/or cooling medium. The secondary system delivers heating, ventilation, cooling, humidification, and/or dehumidification to the occupied space. The systems are independent to a great extent, so several secondary systems may work with a particular primary system. In some cases, however, only one secondary system may be suitable for a particular primary system.

Once subjective analysis has identified one or more HVAC systems (sometimes only one choice remains), detailed quantitative evaluations must be made. All systems considered should provide satisfactory performance to meet the owner's essential goals. The design engineer should provide the owner with specific data on each system to make an informed choice. Consult the following chapters to help narrow the choices:

- Chapter 10 of the 2017 *ASHRAE Handbook—Fundamentals* covers physiological principles, comfort, and health.
- Chapter 19 of the 2017 *ASHRAE Handbook—Fundamentals* covers methods for estimating annual energy costs.
- Chapters 37 to 44 of the 2019 *ASHRAE Handbook—HVAC Applications* address important topics pertinent to the long-term success of building programs.

Other documents and guidelines that should be consulted are ASHRAE standards and guidelines; local, state, and federal guidelines; and special agency requirements (e.g., U.S. General Services Administration [GSA], Food and Drug Administration [FDA], Joint Commission on Accreditation of Healthcare Organizations [JCAHO], Facility Guidelines Institute [FGI], Leadership in Energy and Environmental Design [LEED®]). Additional sources of detailed information are listed in the Bibliography and/or available in the ASHRAE Bookstore (www.ashrae.org/bookstore).